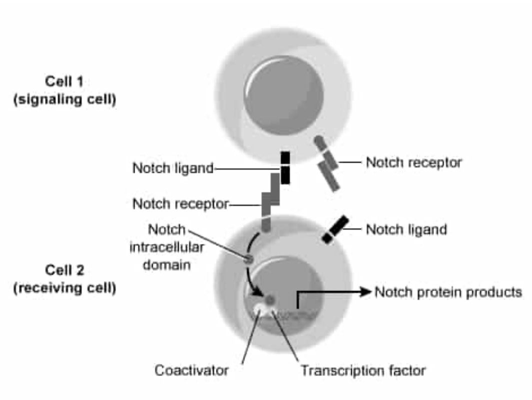
In the figure below, a Notch signaling pathway involving two developing embryonic cells is shown. Notch protein products inhibit the Notch ligand and promote cell lineage commitment in the receiving cell.

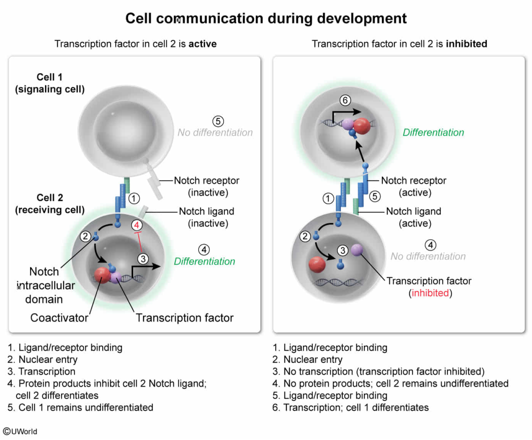


A compound that inhibits transcription factor activity in Cell 2 is most likely to have which of the following effects on development in these cells?

- A. Differentiation will be inhibited in both Cell 1 and Cell 2. [20%]
- B. Differentiation will be induced in both Cell 1 and Cell 2. [4%]
- ✔ C. Differentiation will be induced in Cell 1; differentiation will be inhibited in Cell 2. [61%]
- D. Differentiation will be inhibited in Cell 1; differentiation will be induced in Cell 2. [12%]

Correct
62% answered correctly

Explanation:



Morphogens (eg, Notch) are signaling molecules that change patterns of embryonic cell differentiation in a concentration-dependent manner. Cells that respond to morphogens (due to the presence of receptors) are known as competent cells. After morphogens are released from signaling cells, the molecules diffuse from one region of the embryo to the other, creating a concentration gradient.

Some morphogens are paracrine factors, acting on cells that are close to the signaling cell.

Other morphogens signal via direct contact mechanisms, in which a ligand attached to the plasma membrane of the signaling cell interacts with a receptor on the membrane of an adjacent cell.

The question figure shows the Notch signaling pathway involving two developing embryonic cells: Cell 1 (the signaling cell) and Cell 2 (the receiving cell). The Notch ligand shown on the Cell 1 membrane is interacting with the Notch receptor on Cell 2. This interaction leads to the release of the Notch intracellular domain into the cytoplasm of Cell 2, where it enters the nucleus and interacts with a coactivator and a transcription factor. Subsequently, Notch protein products are produced; the question states that these protein products cause inhibition of the Notch ligand and cell lineage commitment (ie, differentiation) in the receiving cell (ie, Cell 2).

Because the Notch ligand on Cell 2 is inhibited, the Notch receptor on Cell 1 will remain in an inactivated state. Consequently, the Notch intracellular domain will not be released in Cell 1, preventing the production of Notch protein products. This will cause Cell 1 to remain in an undifferentiated state.

If a compound inhibits transcription factor activity in Cell 2, Notch protein products will not be produced. Therefore, **differentiation will be inhibited in Cell 2 (Choices B and D)**. Likewise, without Notch protein products, the Notch ligand located in the Cell 2 membrane will not be inhibited. Consequently, this ligand will be able to interact with the Notch receptor on Cell 1, **inducing differentiation in Cell 1 (Choice A)**.

Educational objective:
Morphogens are signaling molecules that change patterns of embryonic cell differentiation in a concentration-dependent manner. Interfering with the signaling pathways induced by morphogens can change patterns of differentiation in an embryo.

Time spent:0 sec	**QID**:403382
Subject: Biology	**System:** Reproduction